통계열역학 (2011년도 1학기)

2차 과제 정답 (**총 70 점**)

1. **(총 10점, 각 5점)**

(a)

$$\begin{aligned} q_{elec}(T) &= \sum_{i=1} g_{ei} e^{-\beta\epsilon_{ei}} = \sum_{i=1} g_{ei} \exp\left[-\frac{hc\bar{\nu}}{k_B T} \right] \\ &= \sum_{i=1} g_{ei} \exp\left[-\frac{(6.62608\times10^{-34}\ \text{J s})(2.99792458\times10^{10}\ \text{cm s}^{-1})}{(1.38066\times10^{-23}\ \text{J K}^{-1})} \cdot \frac{\bar{\nu}(\text{cm}^{-1})}{T} \right] \\ &= \sum_{i=1} g_{ei} \exp\left[-(1.439\ \text{cm K}^{-1}) \frac{\bar{\nu}(\text{cm}^{-1})}{T} \right]. \end{aligned}$$

위 식에서 g_{ei}는 각 에너지 준위에서 축퇴도이고 ϵ_{ei}는 i 번째 에너지 준위의 에너지이다. 편의상 $\epsilon_{e1} = 0$ 0이라고 한다.

$$\begin{aligned} q(298\ \text{K}) &= 5 + 1e^{-1.439\times4707/298} + 3e^{-1.439\times4751/298} + 5e^{-1.439\times10559/298} = 5.00 \\ q(5000\ \text{K}) &= 5 + 1e^{-1.439\times4707/5000} + 3e^{-1.439\times4751/5000} + 5e^{-1.439\times10559/5000} = 6.26 \end{aligned}$$

(b) Te 원자가 바닥상태 (에너지 준위 1)와 첫 번째 들뜬상태 (에너지 준위 2)에 존재할 확률은 각각

$$p_{ei} = \frac{g_{ei} e^{-\beta\epsilon_{ei}}}{q_{\text{elec}}(T)}.$$

온도	p_1	p_2	p_3	p_4
(i) 298 K	1.0	0.0	0.0	0.0
(ii) 5000 K	0.80	0.040	0.12	0.04

2. **(10점)** 이원자분자 이상기체의 몰엔트로피는 다음과 같이 주어진다.

$$\begin{aligned} \bar{S} &= N_A k_B \ln\left(\frac{V}{\Lambda^3} \left(\frac{e^{5/2}}{N_A} \right) \right) + N_A k_B + N_A k_B \ln\left(\frac{T}{\sigma\Theta_{rot}} \right) \\ &\quad + N_A k_B \left(\frac{\Theta_{vib}/T}{1 - e^{-\Theta_{vib}/T}} - \ln\left(1 - e^{-\Theta_{vib}/T}\right) \right) \\ &= R \ln\left(\frac{V}{\Lambda^3} \left(\frac{e^{5/2}}{N_A} \right) \right) + R + R \ln\left(\frac{T}{\sigma\Theta_{rot}} \right) \\ &\quad + R \left(\frac{\Theta_{vib}/T}{e^{\Theta_{vib}/T} - 1} - \ln\left(1 - e^{-\Theta_{vib}/T}\right) \right) \end{aligned} \tag{1}$$

1 몰의 N^2 기체가 이상기체와 같이 행동한다고 가정하면, 298.15 K, 1 bar에서 몰부피는

$$V = \frac{nRT}{P} = \frac{(1\ \text{mol})(8.31451\ \text{J K mol}^{-1})(298.15\ \text{K})}{(1.000 \times 10^5\ \text{N m}^{-2})} = 0.02479\ \text{m}^3 \tag{2}$$

한편, 열적파장 Λ는

$$\begin{aligned}
\Lambda &= \left(\frac{h^2}{2\pi M k_B T}\right)^{1/2} \\
&= \left(\frac{(6.62608 \times 10^{-34}\ \text{J s})^2}{2\pi(28.0 \times 1.66054 \times 10^{-27}\ \text{kg})(1.38066 \times 10^{-23}\ \text{J K}^{-1})(298.15\text{K})}\right)^{1/2} \\
&= 1.90922 \times 10^{-11}\ \text{m}
\end{aligned} \tag{3}$$

298.15 K에서 $\bar{S}^\circ$는 (1)부터 (3) 식에 의해서

$$\begin{aligned}
\bar{S}^\circ &= (8.31451\ \text{J K}^{-1}\ \text{mol}^{-1}) \times \ln\left(\frac{(0.02479\ \text{m}^3) \times e^{5/2}}{(1.90922 \times 10^{-11}\ \text{m})^3 \times (6.02214 \times 10^{23})}\right) \\
&\quad +(8.31451\ \text{J K}^{-1}\ \text{mol}^{-1}) + (8.31451\ \text{J K}^{-1}\ \text{mol}^{-1}) \times \ln\left(\frac{298.15\ \text{K}}{2 \times 2.88\ \text{K}}\right) \\
&\quad +\frac{(8.31451\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (3398\ \text{K}/298.15\ \text{K})}{e^{3374\ \text{K}/298.15\ \text{K}} - 1} \\
&\quad -(8.31451\ \text{J K}^{-1}\ \text{mol}^{-1}) \times \ln\left(1 - e^{-3374\ \text{K}/298.15\ \text{K}}\right) \\
&= 191.45\ \text{J K}^{-1}\ \text{mol}^{-1}
\end{aligned}$$

3. (**총 20점, 각 10점**)

(a)

$$\frac{\bar{C}_V(T)}{R} = 2.283 + (6.291 \times 10^{-4}\text{K}^{-1})T - (5.0 \times 10^{-10}\text{K}^{-2})T^2. \tag{4}$$

(b)

$$\begin{aligned}
\frac{\bar{C}_V}{R} &= \frac{5}{2} + \left(\frac{\Theta_{vib}}{T}\right)^2 \frac{e^{-\Theta_{vib}/T}}{(1 - e^{-\Theta_{vib}/T})^2} \\
&= \frac{5}{2} + \left(\frac{3374\ \text{K}}{T}\right)^2 \frac{e^{-3374\ \text{K}/T}}{(1 - e^{-3374\ \text{K}/T})^2}
\end{aligned} \tag{5}$$

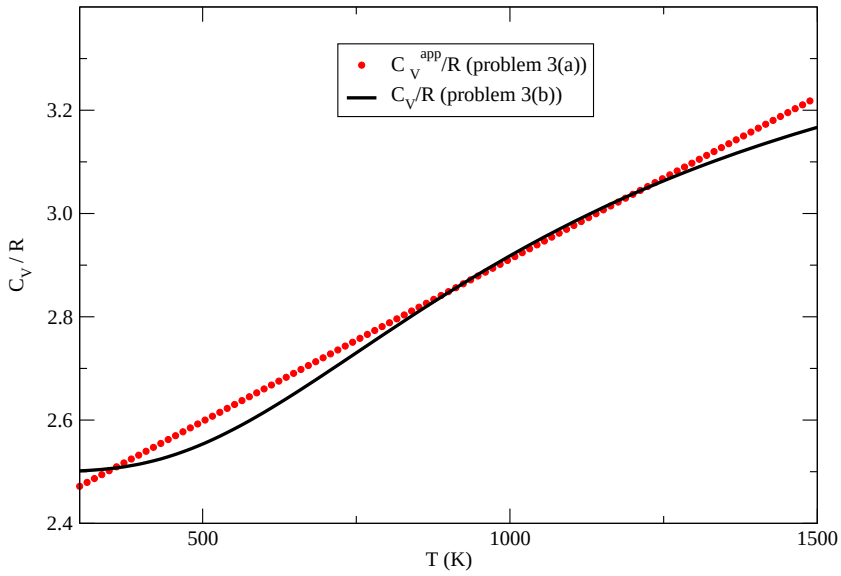


그림 1 – 문제 3

4. (**총 20점, 각 10점**)

(a) $m_{\rm I} = 127$ g/mol $= 127 \times 10^{-3}$kg/$6.02 \times 10^{23} = 2.11 \times 10^{-25}$ kg. $I_2(g)$ 와 $I(g)$ 의 분자분배함수

$$
\begin{aligned}
q_{\rm I_2}/V &= \left(\frac{2\pi m_{I_2} k_B T}{h^2}\right)^{3/2} \cdot \left(\frac{T}{2\Theta_{\rm rot}}\right) \cdot \left(\frac{e^{-\Theta_{\rm vib}/(2T)}}{1-e^{-\Theta_{\rm vib}/T}}\right) \cdot g_{e0}(I_2) e^{D_e/(k_B T)} \\
&= \left(\frac{(2\pi)(2\times 2.11\times 10^{-25}\text{kg})(1.381\times 10^{-23}\ \text{J K}^{-1})(1000\text{K})}{(6.626\times 10^{-34}\ \text{J s})^2}\right)^{3/2} \\
&\quad \times \left(\frac{1000\ \text{K}}{2\times 0.0537\ \text{K}}\right)\left(\frac{e^{-(308\ \text{K})/(2\times 1000\ \text{K})}}{1-e^{-(308\ \text{K})/(1000\ \text{K})}}\right)(1) e^{(150.3\ \text{kJ mol}^{-1})/((1000\ \text{K})(1.381\times 10^{-23}\ \text{J K}^{-1}))}, \\
&= 5.15\times 10^{45}\ \text{m}^{-3} \qquad (6)
\end{aligned}
$$

$$
\begin{aligned}
q_I/V &= \left(\frac{2\pi m_I k_B T}{h^2}\right)^{3/2} V \cdot g_{e0}(I) \\
&= \left(\frac{(2\pi)(2.11\times 10^{-25}\text{kg})(1.381\times 10^{-23}\text{J K}^{-1})(1000\text{K})}{(6.626\times 10^{-34}\text{J s})^2}\right)^{3/2} (4) \\
&= 3.41\times 10^{34}\ \text{m}^{-3} \qquad (7)
\end{aligned}
$$

그러므로 평형상수 $K_p(1000\ \text{K})$ 의 식 :

$$
\begin{aligned}
K_p(1000\ \text{K}) &= (k_B T)\frac{(q_{\rm I}/V)^2}{(q_{\rm I_2}/V)} \\
&= (1.381\times 10^{-23}\ \text{J K}^{-1})(1000\ \text{K})\frac{(3.41\times 10^{34}\ \text{m}^{-3})^2}{(5.15\times 10^{45}\ \text{m}^{-3})} \\
&= 311\ \text{N m}^{-2} = 3.1\times 10^{-3}\ \text{atm} \qquad (8)
\end{aligned}
$$

(b) 그림

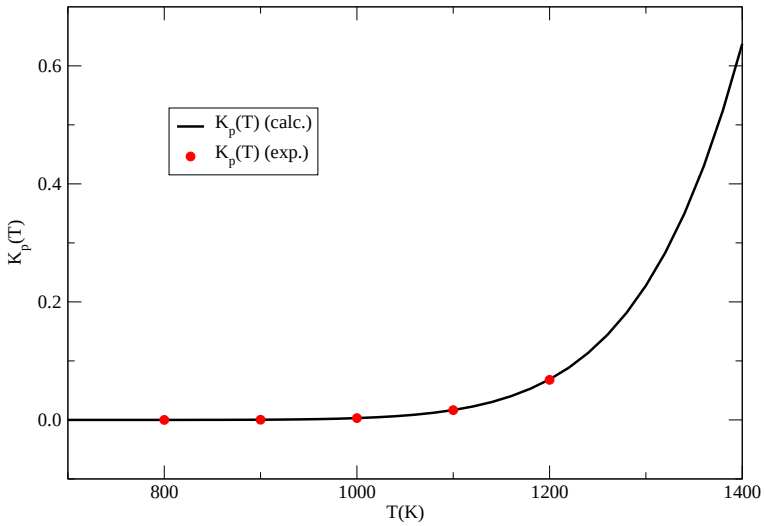


그림 2 – 문제 4(b)

5. (**총 20점, 각 10점**)

(a) $\bar{C}_V^{\text{exp}}/R$에 대한 그래프

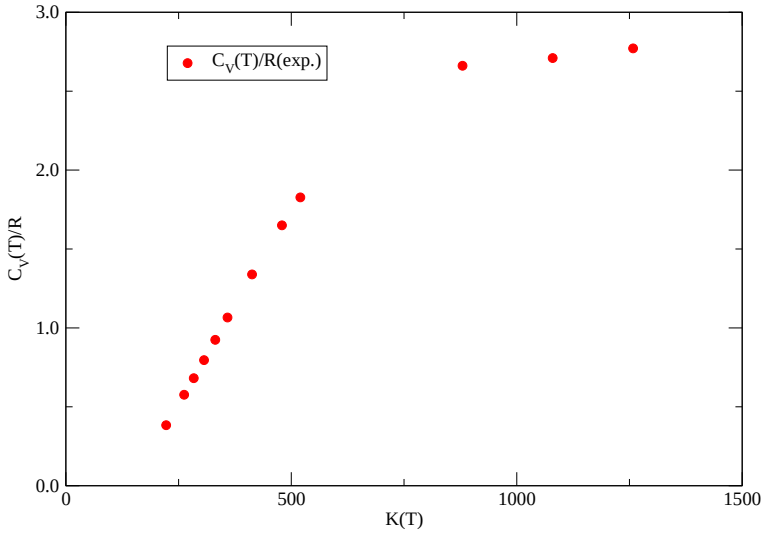


그림 3 – 문제 5(a)

(b) Θ_E(다이아몬드) = 1320 K

$\bar{C}_V^{\exp}/R$와 $\bar{C}_{V,E}/R$에 대한 그래프

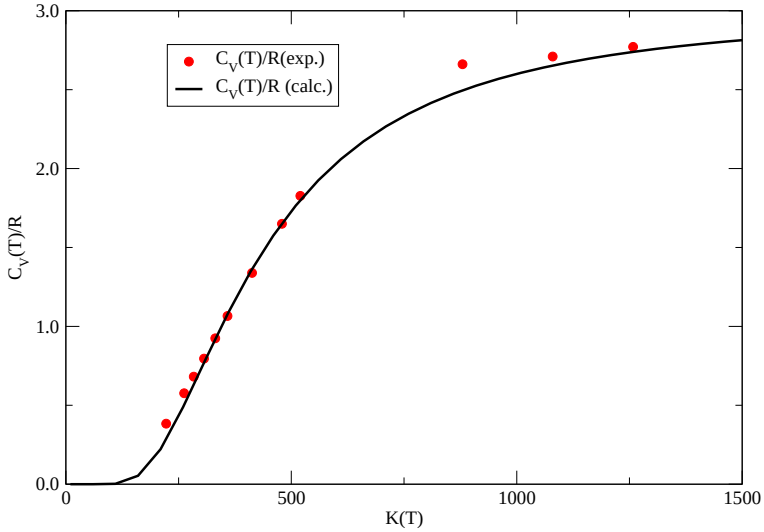


그림 4 – 문제 5(b)